Fundamentals of AI and ML
(CSA2001)

README: AI Travel Planner

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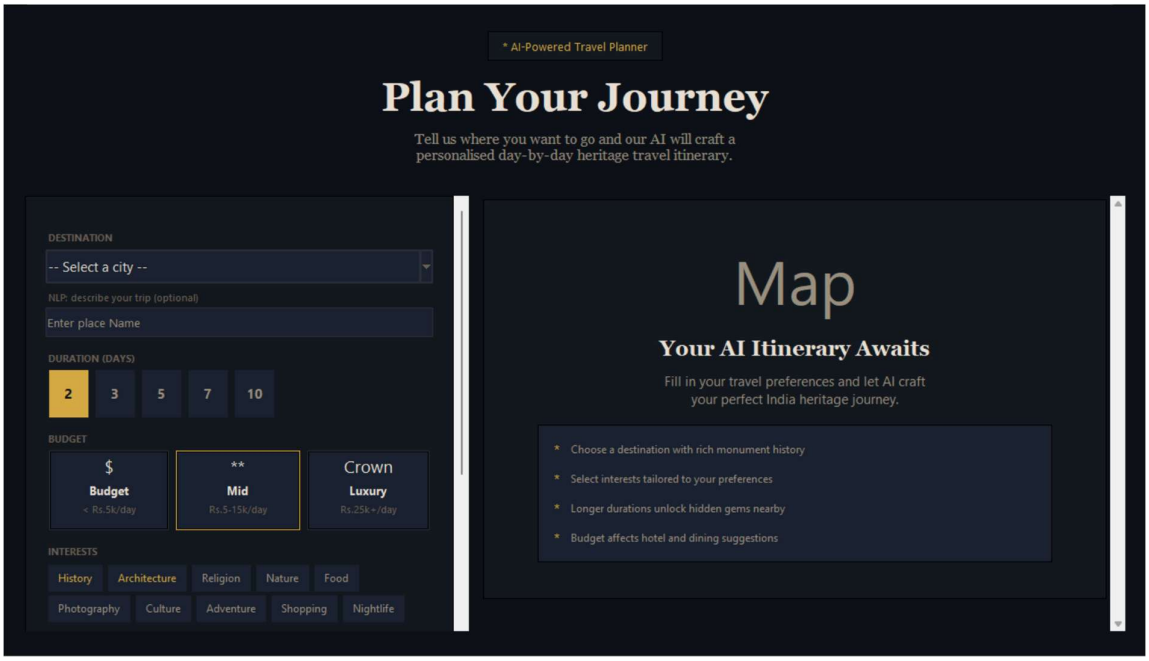
AI Travel Planner is a premium Python-based desktop application designed to simplify the creation of heritage-focused travel itineraries. Using a custom-themed interface and a robust PDF engine, it transforms travel data into structured, timeblocked plans.



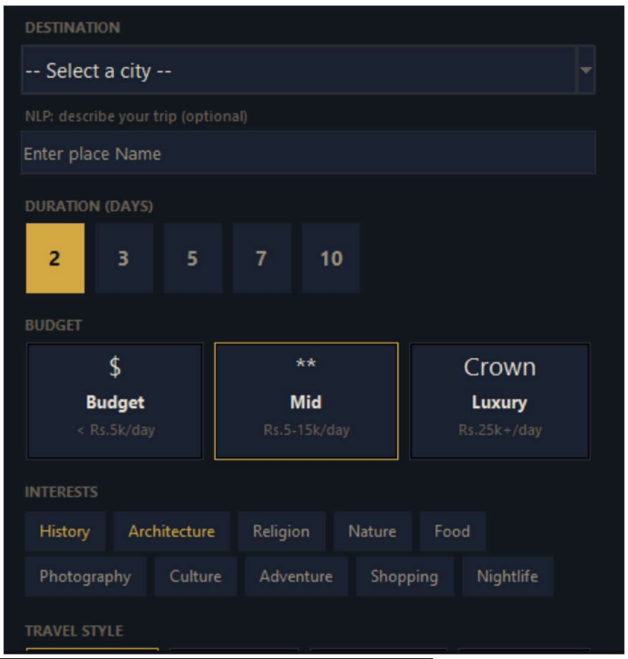
Project Overview

This application caters to travellers seeking cultural depth. It features a curated knowledge base of over 15 major Indian heritage hubs, including **Agra, Delhi, Jaipur, Hampi, Varanasi, etc**.

Key Features

- **Intelligent Scheduling:** Automatically generates detailed itineraries for 2, 3, 5, 7, or 10 days.
- **Time-Blocked Planning:** Daily schedules are divided into **Morning** (7 AM), **Afternoon** (1 PM), and **Evening** (6 PM) slots.
- **Budget Tiering:** Integrated cost estimations for **Budget**, **Mid-range**, and **Luxury** travel styles.

- **Professional PDF Export:** Dynamically creates branded A4 PDF documents using the ReportLab engine.
- **Modern UI:** Features a sleek, dark-themed dashboard with antique gold accents.
- **Simulation Engine:** Provides real-time feedback during processing, such as "Mapping heritage sites by era" and "Optimising travel distances".



Technical Stack

- **Language:** Python 3.x
- **GUI Framework:** Tkinter (Custom themed)
- **PDF Generation:** ReportLab
- **ML Models:** Nearest Neighbours and Linear Regression from **Scikit-Learn**.
- **Concurrency:** Threading (ensures the UI remains responsive during document generation)
- **Utilities:** Webbrowser, Datetime, OS, Tempfile

Installation & Setup

1. **Clone the Repository:** Bash

```
git clone
https://github.com/Divyanshgodgh
ate/Travel-Itinerary
```

2. **Install Dependencies:** Bash

```
pip install reportlab
pip install scikit-learn
pip install numpy
pip install pandas
```

3. **Run the Application:** Bash

```
python Travel.py
```

How to Use

1. **Select Destination:** Choose a city from the dropdown menu (e.g., Delhi, Mysore, or Mumbai).
2. **Set Preferences:** Choose the trip duration and pace (Relaxed, Packed, Focused, or Flexible).
3. **Generate:** Click "Generate Plan" to begin the AI simulation steps.
4. **View/Export:** Once complete, open the professional PDF or share a digital link to view it in a browser.